

UNIT 2: Spectroscopic Techniques & Applications

1. UV-Visible Spectroscopy

UV-Vis spectroscopy deals with the absorption of ultraviolet (200-400 nm) or visible (400-800 nm) radiation, promoting valence electrons from bonding/non-bonding orbitals to antibonding orbitals.

Beer-Lambert Law (Core Formula):

When monochromatic light passes through a homogeneous solution, the rate of decrease of intensity with path length is proportional to light intensity and solution concentration.

$$A = \log_{10}(I_0 / I) = \epsilon \cdot c \cdot l$$

where A = Absorbance, ϵ = Molar absorptivity ($L \text{ mol}^{-1} \text{ cm}^{-1}$), c = Concentration (mol/L), l = Path length (cm).

Key Terminology & Spectral Shifts:

- **Chromophore:** An isolated unsaturated functional group responsible for light absorption (e.g., -C=C- , -C=O , -N=N-).
- **Auxochrome:** A non-absorbing group with lone pairs that intensifies color when attached to a chromophore (e.g., -OH , -NH_2 , -Cl).
- **Bathochromic Shift (Red Shift):** Shift of absorption maximum (λ_{max}) to longer wavelength.
- **Hypsochromic Shift (Blue Shift):** Shift of λ_{max} to shorter wavelength.
- **Hyperchromic Effect:** Increase in absorption intensity (ϵ_{max}).
- **Hypochromic Effect:** Decrease in absorption intensity (ϵ_{max}).

2. Infrared (IR) Spectroscopy

IR radiation (4000 to 400 cm^{-1}) excites molecular vibrations (stretching and bending of chemical bonds). Think of chemical bonds as mechanical springs!

Hooke's Law for Vibrational Frequency:

$$\nu = \frac{1}{2\pi} \sqrt{k / \mu}$$

where k = bond force constant (triple bond > double bond > single bond), μ = reduced mass = $(m_1 m_2) / (m_1 + m_2)$.

Vibrational Modes & Spectral Regions:

- **Stretching Vibrations:** Distance between atoms changes along bond axis (Symmetric vs Asymmetric).
- **Bending Vibrations:** Bond angle changes (In-plane: Scissoring/Rocking; Out-of-plane: Wagging/Twisting).
- **Functional Group Region ($4000 - 1500 \text{ cm}^{-1}$):** Characteristic peaks for -OH ($3200\text{-}3600 \text{ cm}^{-1}$ broad), >C=O (1700 cm^{-1} sharp peak).
- **Fingerprint Region ($1500 - 400 \text{ cm}^{-1}$):** Unique absorption pattern for every organic molecule.

3. ^1H -NMR Spectroscopy (Nuclear Magnetic Resonance)

NMR uses radiofrequency radiation in a strong magnetic field (B_0) to study hydrogen nuclei (protons) in an organic molecule.

Chemical Shift (δ in ppm):

Position of proton absorption relative to reference compound Tetramethylsilane (TMS = 0 ppm).

$$\delta = [(v_{\text{sample}} - v_{\text{TMS}}) / \text{Operating Freq in MHz}] \times 10^6$$

NMR Interpretation Rules:

1. Number of Signals: Gives the number of non-equivalent sets of protons.
2. Spin-Spin Splitting (n+1 Rule): A proton signal splits into (n + 1) peaks, where n = number of neighboring protons on adjacent carbons. (e.g., $-\text{CH}_2-\text{CH}_3$ CH_3 splits CH_2 into triplet $2+1=3$).
3. Peak Area (Integration): Area under peak is proportional to number of protons in that chemical environment.

Structural Elucidation Example (Ethanol $\text{CH}_3\text{CH}_2\text{OH}$):

- CH_3 protons (3H): Splits CH_2 signal Quartet (1:3:3:1) at δ 1.2 ppm.
- CH_2 protons (2H): Split by CH_3 Triplet (1:2:1) at δ 3.7 ppm.
- $-\text{OH}$ proton (1H): Singlet at δ 4.8 ppm.

4. Quick Exam Revision Summary

- Beer-Lambert: $A = \epsilon c l$.
- Chromophore: Light absorbing group ($-\text{C}=\text{C}-$) vs Auxochrome: Color enhancing group ($-\text{OH}$).
- IR Region: Functional ($4000-1500 \text{ cm}^{-1}$) vs Fingerprint ($1500-400 \text{ cm}^{-1}$).
- NMR Reference: TMS at $\delta = 0$ ppm; Peak splitting = (n+1) rule.